



Exam 1 Review Problem Set: Solutions

Differential Equations

Summer 2026

1. Solve the following initial value problem (IVP) on $y = y(x)$, and specify the domain for your solution:

$$\begin{cases} y' = (x \log x)^{-1}, \\ y(e) = -6. \end{cases}$$

Solution:

Here, we notice that this problem is separable, hence we can write:

$$dy = \frac{1}{x \log x} dx,$$

$$\int dy = \int \frac{1}{x \log x} dx.$$

Now, we evaluate the integral by substitution, *i.e.*, $u = \log x$ and $du = dx/x$, which give that:

$$y = \int \frac{1}{u} du = \log |u| + C = \log |\log x| + C.$$

Eventually, we plug in the initial condition, that is $y(e) = -6$, giving us that:

$$-6 = \log |\log e| + C,$$

$$C = -6.$$

Therefore, the solution is:

$$y = \boxed{\log |\log x| - 6}.$$

Here, we note that $\log(-)$ has a valid domain over positive numbers, and the double $\log(-)$ functions enforces that x must be greater than 1, as $\log(0)$ is undefined. Since our initial condition is e , and $e \in (1, \infty)$, the domain of the solution is $\boxed{(1, \infty)}$.

2. Suppose $f(x)$ is non-zero, let an initial value problem be:

$$\begin{cases} \frac{1-y}{x} \cdot \frac{dy}{dx} = \frac{f(x)}{1+y}, \\ y(0) = 0. \end{cases}$$

(a) Show that the differential equation is **not** linear.

For the next two questions, suppose $f(x) = \tan x$.

(b) State, without justification, the open interval(s) in which $f(x)$ is continuous.

(c)* Show that there exists some $\delta > 0$ such that there exists a unique solution $y(x)$ for $x \in (-\delta, \delta)$.

Now, suppose that $f(x)$ is some function, **not** necessarily continuous.

(d) Suppose that the condition in (c) does **not** hold, give three examples in which $f(x)$ could be.

Solution:

(a) *Proof.* We can write the equation as $F(x, y, y') := y' - \frac{xf(x)}{(y+1)(y-1)} = 0$, and since:

$$F(x, (y+1), (y+1)') = y' - \frac{xf(x)}{(y+2)y} \neq 1,$$

so the function is non-linear. □

(b) Here, we should consider that:

$$f(x) = \tan x = \frac{\sin x}{\cos x},$$

so the discontinuities are at when $\cos x = 0$, that is:

$$x \in \left\{ \frac{(2k+1)\pi}{2} : k \in \mathbb{Z} \right\}.$$

Hence, we have the intervals in which $f(x)$ being continuous as:

$$\left\{ \left(\frac{(2k-1)\pi}{2}, \frac{(2k+1)\pi}{2} \right) : k \in \mathbb{Z} \right\}.$$

(c) *Proof.* Here, we want to write our equation in the standard form and obtain that:

$$y' := f(t, y) = \frac{x \tan x}{(y+1)(y-1)}, \quad \frac{\partial f(t, y)}{\partial y} = -\frac{x \tan x \cdot 2y}{(y^2-1)^2}.$$

Clear, we note the discontinuities of y at $y = \pm 1$, and x demonstrated as above, thus we can form a rectangle $Q = (-\pi/2, \pi/2) \times (-1, 1)$ in which the initial condition $(0, 0) \in Q$ and $f(t, y)$ with $\partial_y f(t, y)$ are continuous on the interval. By the *existence and uniqueness theorem for non-linear case*, we know that there exists some δ such that there is a unique solution for $-\delta < x < \delta$. □

(d) If the condition in (c) does not hold, by contraposition, this implies that continuity must fail, i.e., $xf(x)$ must be discontinuous at $x = 0$. Hence, some examples could be:

$$f(x) = \frac{1}{x^2}, \text{ or } \log x, \text{ or } \csc x, \text{ or } \chi_{\{0\}}(x) \text{ etc.}$$

3. Solve for the general solution to the following ODEs with $y = y(t)$:

(a) $2y' + y = 3t.$

(b) $y' + \log(t)y = t^{-t}.$

Solution:

(a) Here, we first convert the equation to standard form, *i.e.*:

$$y' + \frac{1}{2}y = \frac{3}{2}t.$$

Hence, with $p(t) = 1/2$, the integration factor must be:

$$\mu(t) = \exp\left(\int_0^t p(s)ds\right) = \exp\left(\int_0^t \frac{1}{2}ds\right) = \exp\left(\frac{1}{2}t\right).$$

Now, we multiply the integration factor on both sides, giving that:

$$\begin{aligned} y'e^{t/2} + \frac{1}{2}ye^{t/2} &= \frac{3}{2}te^{t/2}, \\ \frac{d}{dt}\left[e^{t/2}y\right] &= \frac{3}{2}te^{t/2}, \\ e^{t/2}y &= \frac{3}{2}\int te^{t/2}dt = \frac{3}{2}\left[2te^{t/2} - \int 2e^{t/2}\right] \\ &= \frac{3}{2}\left[2te^{t/2} - 4e^{t/2} + C\right] = 3te^{t/2} - 6e^{t/2} + \tilde{C}, \\ y &= \boxed{\tilde{C}e^{-t/2} + 3t - 6}. \end{aligned}$$

(b) Again, we find the integration factor as:

$$\mu(t) = \exp\left(\int_e^t \log s ds\right).$$

To find the antiderivative of $\log(-)$, we have:

$$\int \log s ds = \int 1 \cdot \log s ds = s \log s - \int s \cdot \frac{1}{s} ds = s \log s - \int ds = s \log s - s + C.$$

Therefore, the integrating factor is:

$$\mu(t) = \exp(t \log t - t) = \frac{\exp(t \log t)}{e^t} = \frac{t^t}{e^t}.$$

Hence, we multiply the integration factor on both sides, giving that:

$$\begin{aligned} \frac{t^t}{e^t} + \frac{t^t}{e^t} \log ty &= e^{-t}, \\ \frac{d}{dt}\left[\frac{t^t}{e^t}y\right] &= e^{-t}, \\ \frac{t^t}{e^t}y &= \int e^{-t}dt = -e^{-t} + C, \\ y &= \boxed{-t^{-t} + Ct^{-t}e^t}. \end{aligned}$$

4. Solve the following initial value problem (IVP) on $y = y(x)$, and specify the domain for your solution:

$$\begin{cases} y' = y(y+1), \\ y(0) = 1. \end{cases}$$

Solution: Now, we trivially separate the problem as:

$$\frac{dy}{y(y+1)} = dx.$$

Then, we use the partial fraction to obtain that:

$$\left(\frac{1}{y} - \frac{1}{y+1} \right) dy = dx.$$

Now, we can integrate both side to obtain that:

$$\int \left(\frac{1}{y} - \frac{1}{y+1} \right) dy = \int dx,$$

$$\log |y| - \log |y+1| = x + C,$$

$$\log \left| \frac{y}{y+1} \right| = x + C,$$

$$\frac{y}{y+1} = \tilde{C}e^x.$$

Now, we consider the left hand side as:

$$\frac{y}{y+1} = \frac{y+1}{y+1} - \frac{1}{y+1} = 1 - \frac{1}{y+1},$$

which allows us to rewrite the equation as:

$$\frac{1}{y+1} = 1 - \tilde{C}e^x,$$

$$y = \frac{1}{1 - \tilde{C}e^x} - 1 = \frac{\tilde{C}e^x}{1 - \tilde{C}e^x}.$$

By plugging in the initial condition, we trivially have:

$$1 = \frac{\tilde{C}}{1 - \tilde{C}} \implies \tilde{C} = \frac{1}{2}.$$

Hence, the solution is:

$$y = \frac{e^x}{2 - e^x}.$$

For the domain of our solution, we note that e^x is continuous, but $2 - e^x$ could cause a zero denominator at $x = \log 2$. Note that $\log 2 > 0$, so the domain of the solution is $(-\infty, \log 2)$.

5. Given an initial value problem:

$$\begin{cases} \frac{dy}{dt} - \frac{3}{2}y = 3t + 2e^t, \\ y(0) = y_0. \end{cases}$$

- Find the integrating factor $\mu(t)$.
- Solve for the particular solution for the initial value problem.
- Discuss the behavior of the solution as $t \rightarrow \infty$ for different cases of y_0 .

Solution:

(a) As instructed, we look for the integrating factor as:

$$\mu(t) = \exp\left(\int_0^t -\frac{3}{2}ds\right) = \exp\left(-\frac{3}{2}t\right).$$

(b) With the integrating factor, we multiply both sides by $\mu(t)$ to obtain that:

$$y'e^{-3t/2} - \frac{3}{2}ye^{-3t/2} = 3te^{-3t/2} + 2e^te^{-3t/2}.$$

Clearly, we observe that the left hand side is the derivative after product rule for $ye^{-3t/2}$ and the right hand side can be simplified as:

$$\frac{d}{dt} [ye^{-3t/2}] = 3te^{-3t/2} + 2e^{-t/2}.$$

Therefore, we have turned this into an integration problem, so we do the respective integrations, giving us that:

$$\begin{aligned} ye^{-3t/2} &= \int 3te^{-3t/2}dt + \int 2e^{-t/2}dt \\ &= -2te^{-3t/2} + 2 \int e^{-3t/2}dt - 4e^{-t/2} + C \\ &= -2te^{-3t/2} - \frac{4}{3}e^{-3t/2} - 4e^{-t/2} + C. \end{aligned}$$

Then, we divide both sides by $e^{-3t/2}$ to get the general solution:

$$y(t) = -2t - \frac{4}{3} - 4e^t + Ce^{3t/2}.$$

Given the initial condition, we have that:

$$y_0 = 0 - \frac{4}{3} - 4 + C,$$

which implies $C = 16/3 + y_0$, leading to the particular solution that:

$$y(t) = \left[-2t - \frac{4}{3} - 4e^t + \left(\frac{16}{3} + y_0\right)e^{3t/2}\right].$$

(c) We observe that:

$$\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} \left[-2t - \frac{4}{3} - 4e^t + \left(\frac{16}{3} + y_0\right)e^{3t/2}\right].$$

Note that the important terms are e^t and $e^{3t/2}$, we need to care the critical value $-16/3$:

- when $y_0 > -16/3$, $y(t) \rightarrow \infty$ when $t \rightarrow \infty$,
- when $y_0 \leq -16/3$, $y(t) \rightarrow -\infty$ when $t \rightarrow \infty$.

6. An autonomous differential equation is given as follows:

$$\frac{dy}{dt} = 4y^3 - 12y^2 + 9y - 2 \quad \text{where } t \geq 0 \text{ and } y \geq 0.$$

Draw a phase portrait and sketch a few solutions with different initial conditions.

Solution:

Recall from Pre-Calculus (or Algebra) the following *Rational root test*:

Theorem: Rational Root Test. Let the polynomial:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = 0$$

have integer coefficients $a_i \in \mathbb{Z}$ and $a_0, a_n \neq 0$, then any rational root $r = p/q$ such that $p, q \in \mathbb{Z}$ and $\gcd(p, q) = 1$ satisfies that $p|a_0$ and $q|a_n$. $\lrcorner$

From the theorem, we can note that if the equation has a rational root, it must be one of:

$$r = \pm 1, \pm 2, \pm \frac{1}{2}, \pm \frac{1}{4}.$$

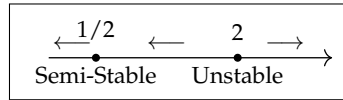
By plugging in, one should notice that $y = 2$ is a root (one might also notice $1/2$ is a root as well, but we will get the step slowly), so we can apply the long division (dividing $y - 2$) to obtain that:

$$\frac{4y^3 - 12y^2 + 9y - 2}{y - 2} = 4y^2 - 4y + 1.$$

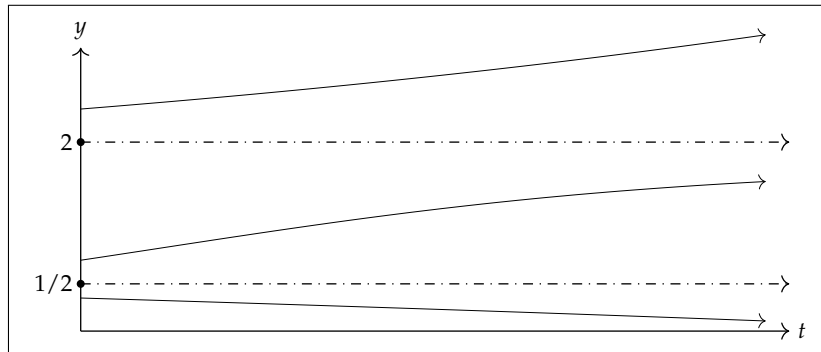
Clear, we can notice that the right hand side is a perfect square (else, you could use the quadratic formula) that:

$$4y^2 - 4y + 1 = (2y - 1)^2.$$

Thus, we now know that the roots are 2 and $1/2$ (multiplicity 2). Hence, the phase portrait is:



Correspondingly, we can sketch a few solutions (not necessarily in scale):



Note that the **Theorem** can also be generalized into the following manner (in ring theory):

Theorem: Rational Root Theorem. Let R be UFD, and polynomial:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in R[x],$$

and let $r = p/q \in K(R)$ be a root of f with $p, q \in R$ and $\gcd(p, q) = 1$, then $p|a_0$ and $q|a_n$. $\lrcorner$

The proofs of the **Theorems** left as exercises to diligent readers.

7. Draw the phase line and determine the stability of each equilibrium for the following autonomous differential equations:

(a) $y' = y^4 - 3y^3 + 2y^2.$

(b) $y' = y^{2025} - 1.$

Solution:

(a) First, we need to factor the right hand side polynomial as:

$$y^4 - 3y^3 + 2y^2 = y^2(y^2 - 3y + 2) = y^2(y - 1)(y - 2),$$

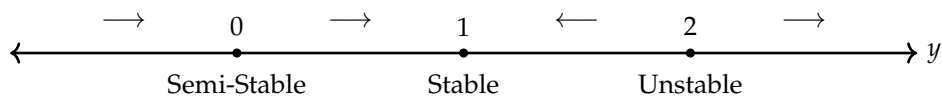
so, we can trivially note the roots as:

$$y = 0 \text{ with multiplicity } 2, y = 1, \text{ and } y = 2.$$

Sophisticated readers shall notice that this polynomial has a positive leading coefficient, hence it approaches $+\infty$ when $y \rightarrow \infty$, hence the arrows can be easily determined.

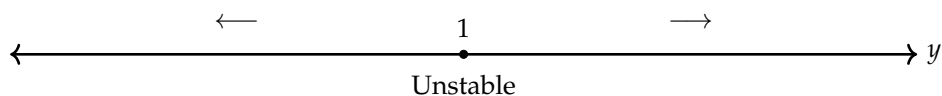
Otherwise, readers can plug in a value within each intervals, such as $y = 3$ for $y > 2$, $y = 3/2$ for $1 < y < 2$, etc., which should work equivalently.

Hence, we should expect a graph as follows:



The stability is given by the directions of the arrows.

(b) Here, readers shall realize that the right hand side polynomial is monotonic $((y^{2025} - 1)' \geq 0)$, so the only real root is at $y = 1$, and since the polynomial has positive leading coefficient and odd order, it shall approach $\mp\infty$ as $y \rightarrow \mp\infty$, so the phase line is:



The stability is given by the directions of the arrows.

8. Let a differential equation be defined as follows:

$$\frac{dy}{dx} = e^{2x} + y - 1.$$

- (a) What is the integrating factor ($\mu(x)$) for the equation? Solve for the general solution.
- (b) Is the equation *exact*? If not, make it exact, then find the general solution.
- (c) Do solutions from part (a) and (b) agree?

Solutions:

- (a) First, we write the equation in standard form, that is $y' - y = e^{2x} - 1$. Hence, with $p(x) = -1$, the integrating factor is:

$$\mu(x) = \exp\left(\int_0^x p(s)ds\right) = \exp\left(\int_0^x (-1)ds\right) = \boxed{\exp(-x)}.$$

Then, we multiply the integrating factors on both ends to obtain:

$$\begin{aligned}\frac{d}{dx} [ye^{-x}] &= e^x - e^{-x}, \\ ye^{-x} &= \int (e^x - e^{-x}) dx = e^x + e^{-x} + C, \\ y &= \boxed{Ce^x + e^{2x} + 1}.\end{aligned}$$

- (b) Note that for exactness, we write the equation as:

$$\underbrace{(-e^{2x} - y + 1)}_{M(x,y)} + \underbrace{(1)}_{N(x,y)} \frac{dy}{dx} = 0,$$

meaning that their partial derivatives are, respectively:

$$\partial_y M(x, y) = -1 \text{ and } \partial_x N(x, y) = 0,$$

and since they are different, the equation is not exact.

Thus, we look for the integrating factor, *i.e.*:

$$\mu(t) = \exp\left(\int_0^x \frac{\partial_y M(x, y) - \partial_x N(x, y)}{N(x, y)}\right) = \exp\left(\int_0^x \frac{-1 - 0}{1} ds\right) = \exp(-x).$$

Now, we multiply e^{-x} on both sides, giving us that:

$$\underbrace{(-e^x - ye^{-x} + e^{-x})}_{\tilde{M}(x,y)} + \underbrace{(e^{-x})}_{\tilde{N}(x,y)} \frac{dy}{dx} = 0.$$

Now, the equation should be exact. *We leave the check to the readers as an exercise.*

To get the solution, we first integrate $\tilde{M}(x, y)$ with respect to x , that is:

$$\varphi(x, y) = \int (-e^x - ye^{-x} + e^{-x}) dx = -e^x + ye^{-x} - e^{-x} + h(y).$$

Now, taking the derivative with respect to y gives:

$$\partial_y \varphi(x, y) = e^{-x} + h'(y) = e^{-x},$$

which pushes $h(y)$ to be constant, hence we have solution:

$$\varphi(x, y) = \boxed{-e^x + ye^{-x} - e^{-x} = C}.$$

- (c) The solutions agree by simple arithmetic deductions.

9.* This brief digression to “differential forms” aims for the following goals:

- Legitimizing $\frac{\partial y}{\partial x} = \frac{f(x)}{g(y)} \iff g(y)dy = f(x)dx$ via the differential operator d .
- Get the foundation of *exactness* for certain differential equation relationship.

First, consider variables $x_1, x_2, \dots, x_n$, we may define the wedge product ($\wedge$) to connect any two variables satisfying that:

$$x_i \wedge x_j = -x_j \wedge x_i \text{ for all } 1 \leq i, j \leq n.$$

(a) Show that $x_i \wedge x_i = 0$ for $1 \leq i \leq n$.

Now, given any smooth function f , we define the differential operator (d) as:

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

(b) Suppose $y(x) = e^x$, find dy .

(c) Now, suppose that $\frac{\partial y}{\partial x} = \frac{f(x)}{g(y)}$, can you express dy in terms of the differential form of x .

Note: Since we have just one variable, we have $dy/dx = \partial y / \partial x$, leading to our first goal.

Furthermore, we can apply the differential operator over differential forms with wedge products already. Suppose:

$$\omega = \sum_{i_1, \dots, i_k} f_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

we may have the differential of ω as:

$$d\omega = \sum_{i_1, \dots, i_k} (df_{i_1, \dots, i_k}) dx_{i_1} \wedge \dots \wedge dx_{i_k}.$$

(d) Suppose x, y are the variables, and $\omega = 2xy^2dx + 2x^2ydy$, show that $d\omega = 0$.

This then relates to a concept called *exactness* in differential equations. Consider the equation:

$$\frac{dy}{dx} + \frac{F(x, y)}{G(x, y)} = 0,$$

we can rewrite it as $F(x, y)dx + G(x, y)dy = 0$. Exactness enforces that:

$$\frac{\partial F}{\partial y} = \frac{\partial G}{\partial x}.$$

Similarly, exactness is considering finding a solution $f(x, y) = c$ such that $F = \frac{\partial f}{\partial x}$ and $G = \frac{\partial f}{\partial y}$.

(e) Show that $df = F(x, y)dx + G(x, y)dy$ and exactness is equivalently $d(df) = 0$.

Note: This implies that the differential equation in part (d) satisfies *exactness*.

Solution:

(a) *Proof.* Since we have:

$$x_i \wedge x_i = -x_i \wedge x_i,$$

we must have $x_i \wedge x_i = 0$. □

(b) By the given differential operator:

$$dy = \frac{\partial y}{\partial x} dx = \boxed{e^x} dx.$$

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(c) Then, we have:

$$dy = \frac{\partial y}{\partial x} dx = \boxed{\frac{f(x)}{g(y)} dx}.$$

Hence, we justify the separation of the variables as $g(y)dy = f(x)dx$.

(d) *Proof.* As instructed, we have:

$$\begin{aligned} d\omega &= \frac{\partial}{\partial x}(2xy^2)dx \wedge dx + \frac{\partial}{\partial y}(2xy^2)dy \wedge dx + \frac{\partial}{\partial x}(2x^2y)dx \wedge dy + \frac{\partial}{\partial y}(2x^2y)dy \wedge dy \\ &= 0 + 4xydy \wedge dx + 4xydx \wedge dy + 0 = -4xydx \wedge dy + 4xydx \wedge dy = 0, \end{aligned}$$

as desired. $\square$

(e) *Proof.* First, we have that:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = Fdx + Gdy.$$

Then, in terms of the exactness relationship, we have:

$$\begin{aligned} \frac{\partial F}{\partial y} = \frac{\partial G}{\partial x} &\iff \frac{\partial F}{\partial y}dy \wedge dx = -\frac{\partial G}{\partial x}dx \wedge dy \\ &\iff \frac{\partial F}{\partial y}dy \wedge dx + \frac{\partial G}{\partial x}dx \wedge dy = 0 \\ &\iff \frac{\partial F}{\partial x}dx \wedge dx + \frac{\partial F}{\partial y}dy \wedge dx + \frac{\partial G}{\partial x}dx \wedge dy + \frac{\partial G}{\partial y}dy \wedge dy = 0 \\ &\iff d(df) = 0. \end{aligned}$$

Hence, we have shown that the exactness is exactly that the differential form satisfies that $d(df) = 0$. $\square$

In fact, for any smooth function f , we have $d(df) = 0$, which is the equivalent of the conclusion such that mixed partials are equal. We invite capable readers to investigate that $d^2 := d \circ d = 0$ for all smooth function f . Additionally, people with experiences in vector calculus could investigate the following *commutative diagram*.

$$\begin{array}{ccccccc} \Omega^0(\mathbb{R}^3) & \xrightarrow{d(-)} & \Omega^1(\mathbb{R}^3) & \xrightarrow{d(-)} & \Omega^2(\mathbb{R}^3) & \xrightarrow{d(-)} & \Omega^3(\mathbb{R}^3) \\ \downarrow \cong & & \downarrow \cong & & \downarrow \cong & & \downarrow \cong \\ \mathcal{C}^\infty(\mathbb{R}^3, \mathbb{R}) & \xrightarrow{\nabla(-)} & \mathcal{C}^\infty(\mathbb{R}^3, \mathbb{R}^3) & \xrightarrow{\nabla \times (-)} & \mathcal{C}^\infty(\mathbb{R}^3, \mathbb{R}^3) & \xrightarrow{\nabla \cdot (-)} & \mathcal{C}^\infty(\mathbb{R}^3, \mathbb{R}) \end{array}$$

The above are respectively 0-form, 1-form, 2-form, and 3-form (with 0, 1, 2, or, 3 $\wedge$'s in the differential form) and the below are smooth functions mapping in respective Euclidean spaces.

- 10.* Determine if the following differential equation is exact. If not, find the integrating factor to make it exact. Then, solve for its general solution;

$$y'(x) = e^{2x} + y(x) - 1.$$

Solution:

First, we write the equation in the general form:

$$\frac{dy}{dx} + (1 - e^{2x} - y) = 0.$$

Now, we take the partial derivatives to obtain that:

$$\frac{\partial}{\partial y}[1 - e^{2x} - y] = -1,$$

$$\frac{\partial}{\partial x}[1] = 0.$$

Notice that the mixed partials are not the same, the equation is not exact.

Here, we choose the integrating factor as:

$$\begin{aligned} \mu(x) &= \exp \left(\int_0^x \frac{\frac{\partial}{\partial y}[1 - e^{2s} - y] - \frac{\partial}{\partial s}[1]}{1} ds \right) \\ &= \exp \left(\int_0^x -ds \right) = \exp(-x). \end{aligned}$$

Therefore, our equation becomes:

$$(e^{-x}) \frac{dy}{dx} + (e^{-x} - e^x - ye^{-x}) = 0.$$

After multiplying the integrating factor, it would be exact. *We leave the repetitive check as an exercise to the readers.*

Now, we can integrate to find the solution with a $h(y)$ as function:

$$\varphi(x, y) = \int (e^{-x} - e^x - ye^{-x}) dx = -e^{-x} - e^x + ye^{-x} + h(y).$$

By taking the partial derivative with respect to y , we have:

$$\partial_y \varphi(x, y) = e^{-x} + h'(y),$$

which leads to the conclusion that $h'(y) = 0$ so $h(y) = C$.

Then, we can conclude that the solution is now:

$$\varphi(x, y) = -e^{-x} - e^x + ye^{-x} + C = 0,$$

which is equivalently:

$$y(x) = \boxed{\widetilde{C}e^x + 1 + e^{2x}}.$$

11. For the first-order autonomous ODE:

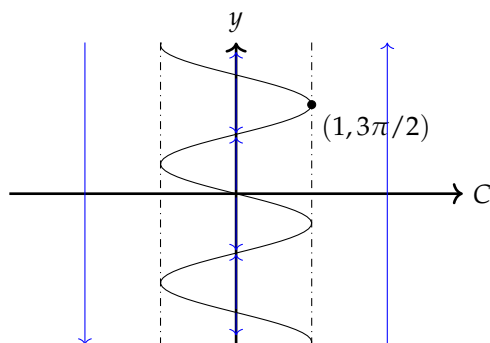
$$\frac{dy}{dt} = \sin y + C,$$

where $C \in \mathbb{R}$ is a parameter. Determine any and all bifurcation values for the parameter C and sketch a bifurcation diagram.

Solution:

It is not hard to observe that $\sin y$ will intersect the axis infinitely many times, while $\sin(\mathbb{R}) = [-1, 1]$, one shall then realize that the bifurcation value would be ± 1 , since when $C > 1$ or $C < -1$, there will be no equilibriums at all.

Therefore, the bifurcation diagram can be illustrated as:



12. Let a first order IVP on $y := y(t)$ be defined as follows:

$$\begin{cases} y' = \frac{2}{t}y, \\ y(1) = 1. \end{cases}$$

- (a) Find the solution to the above initial value problem.
 (b) Recall the theorem on existence and uniqueness, as follows:

For an IVP in simple form:

$$\begin{cases} \frac{dy}{dt} = a(t)y + b(t), \\ y(t_0) = y_0. \end{cases}$$

For some $I = (\alpha, \beta) \ni t_0$, if $a(t)$ and $b(t)$ are continuous on the interval I . Then, there exists a unique solution to the IVP on the interval I .

Show that the IVP in this problem does not satisfy the condition for the existence and uniqueness theorem for $\mathbb{R}$.

- (c) Does the above example violates the existence and uniqueness theorem? Why?

Solution:

- (a) This problem is clearly separable, we may compute:

$$\begin{aligned} \frac{dy}{y} &= 2 \frac{dt}{t} \\ \int \frac{dy}{y} &= 2 \int \frac{dt}{t} \\ \log |y| &= 2 \log |t| + C \\ y &= \tilde{C}t^2. \end{aligned}$$

Note that the initial condition enforces that $y(1) = 1$, so the solution is just:

$$y = t^2.$$

- (b) Note that $a(t) = 2/t$, which is not continuous over $(-\infty, 0) \cup (0, \infty)$, then the theorem does not guarantee the existence and uniqueness of a solution over $\mathbb{R}$.
 (c) This is not a violation since the converse of the theorem is not necessarily true. In propositional logic, if A implies B (written as $A \implies B$), the converse (B implies A , written as $B \implies A$) is not necessarily true. Hence, we can still have a solution that is unique over $\mathbb{R}$.

13. Let a differential equation be defined as:

$$\frac{dy}{dt} = t - y \text{ and } y(0) = 0.$$

Use Euler's Method with step size $h = 1$ to approximate $y(5)$.

Solution:

With $y(0) = 0$, we have $y'(0) = 0 - 0 = 0$. We do the following steps:

- We approximate $y(1) \approx y(0) + 1 \cdot y'(0) = 0$, then we have $y'(1) \approx 1 - 0 = 1$.
- We approximate $y(2) \approx y(1) + 1 \cdot y'(1) \approx 1$, then we have $y'(2) \approx 2 - 1 = 1$.
- We approximate $y(3) \approx y(2) + 1 \cdot y'(2) \approx 2$, then we have $y'(3) \approx 3 - 2 = 1$.
- We approximate $y(4) \approx y(3) + 1 \cdot y'(3) \approx 3$, then we have $y'(4) \approx 4 - 3 = 1$.
- We approximate $y(5) \approx y(4) + 1 \cdot y'(4) \approx 4$.

Then, we have approximated that:

$$y(5) \approx \boxed{4}.$$

14. For the first-order autonomous ODE:

$$\frac{dx}{dt} = x^2 - 2x + c,$$

with parameter $c \in \mathbb{R}$, do the following:

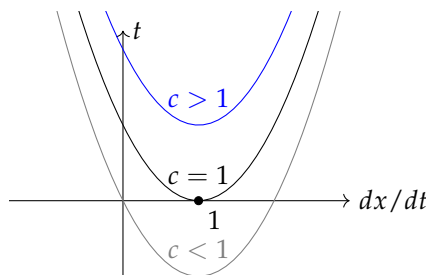
- Sketch all of the qualitatively different graphs of $f(x) = x^2 - 2x + c$, as c is varied.
- Determine any and all bifurcation values for the parameter c .
- Sketch a bifurcation diagram for this ODE.

Solutions:

(a) Given the right hand side, we want to find its critical point, *i.e.*, $x^2 - 2x + c = 0$, that is:

$$x = \frac{2 \pm \sqrt{4 - 4c}}{2} = 1 \pm \sqrt{1 - c}.$$

Graphically, we may draw the diagram as:



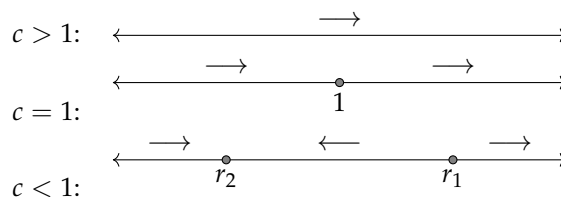
(b) Then, we find the bifurcation value, that is the critical points equivalent to each other, *i.e.*:

$$1 + \sqrt{1 - c} = 1 - \sqrt{1 - c}$$

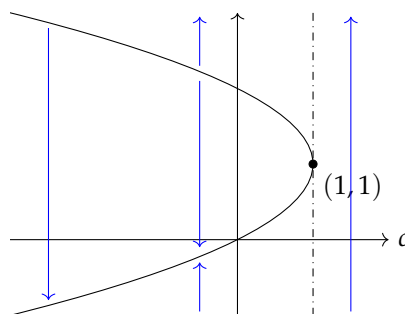
$$c = \boxed{1}.$$

Diligent readers might also notice that this is the value c such that $\Delta = 0$.

(c) When $c > 1$, $dx/dt > 0$. When $c = 1$, the two roots are both 1. When $c < 1$, the $dx/dt > 0$ when larger than the larger root or smaller than the smaller roots, hence the phase diagrams are:



Thus, the bifurcation diagram is:



15. Carbon-14, a radioactive isotope of carbon, is an effective tool in dating the age of organic compounds, as it decays with a relatively long period. Let $Q(t)$ denote the amount of carbon-14 at time t , we suppose that the decay of $Q(t)$ satisfies the following differential equation:

$$\frac{dQ}{dt} = -\lambda Q \text{ where } \lambda \text{ is the rate of decay constant.}$$

- (a) Let the half-life of carbon-14 be τ , find the rate of decay, λ .
 (b) Suppose that a piece of remain is discovered to have 10% of the original amount of carbon-14, find the age of the remain in terms of τ .

Solutions:

- (a) Note that the differential equation is separable, hence:

$$\begin{aligned} \frac{dQ}{Q} &= -\lambda dt, \\ \int \frac{dQ}{Q} &= -\int \lambda dt, \\ \log |Q| &= -\lambda t + C, \\ Q &= \tilde{C}e^{-\lambda t}. \end{aligned}$$

Here, we assume $Q = Q_0$ at $t = t_0$, then we have $Q = Q_0/2$ when $t = t_0 + \tau$, so:

$$\frac{1}{2} = e^{-\lambda\tau},$$

which deduces to:

$$\lambda = -\frac{1}{\tau} \log \left(\frac{1}{2} \right) = \boxed{\frac{\log 2}{\tau}}.$$

- (b) If there are only 10% of remain, we suppose that we have $Q = Q_0$ at $t = 0$, and have $Q = Q_0/10$ at $t = t_0$, hence giving that:

$$\frac{Q_0}{10} = Q_0 \exp(-\lambda t_0) = Q_0 \exp\left(-\frac{\log(2)t_0}{\tau}\right).$$

Thus, we obtain that:

$$\frac{1}{10} = \exp\left(-\frac{\log(2)t_0}{\tau}\right),$$

and by solving for t_0 , we obtain:

$$t_0 = -\frac{\tau}{\log 2} \log \left(\frac{1}{10} \right) = \boxed{\frac{\log 10}{\log 2} \tau}.$$

16.* Let an initial value problem be defined as follows:

$$\begin{cases} (12x^4 + 5x^2 + 6) \frac{dy}{dx} - (x^2 \sin(x) + x^3)y = 0, \\ y(0) = 1. \end{cases}$$

Show that the solution to the above initial value problem is symmetric about $x = 0$.

Solution:

If you were attempting to solve this problem by integrating factor or exactness, you are on the wrong track. The functions are deliberately chosen so that these operations will be hardly possible.

However, this does not necessarily mean that it is not possible to prove without solving the solution out, one shall utilize the existence and uniqueness theorem to proceed.

Proof. Now, we first observe that when we rewrite the problem, we have:

$$y' = \frac{x^2 \sin(x) + x^3}{12x^4 + 5x^2 + 6} y,$$

where we clearly notice that the numerator and denominator are composed of continuous function while the denominator is positive, so we know that it is continuous over $\mathbb{R}$, so the initial value problem exhibits a unique solution.

Now, suppose $y(x)$ is a solution of the above IVP, we want to show that $\tilde{y}(x) := y(-x)$ is also a solution to the above IVP.

Clearly, we have:

$$\tilde{y}(0) = y(-0) = y(0) = 1,$$

so the initial condition is satisfied, so we are left to check the differential equation. By chain rule, we have:

$$\frac{d\tilde{y}}{dx}(x) = \frac{dy}{dx}(-x) \cdot \frac{d}{dx}[-x] = -\tilde{y}'(x).$$

With the first equation and y being a solution, we can make all x into $-x$ to obtain that:

$$(12(-x)^4 + 5(-x)^2 + 6) \frac{dy(-x)}{dx} - ((-x)^2 \sin(-x) + (-x)^3)\tilde{y} = 0,$$

and if we organize the left hand side, we have:

$$(12x^4 + 5x^2 + 6)\tilde{y}' - (x^2 \sin(x) + x^3)\tilde{y} = 0,$$

so $\tilde{y}$ is clearly another solution to the IVP, so by uniqueness, we must have $\tilde{y}(x) = y(x)$, or namely $y(-x) = y(x)$, so the solution must be symmetric about $x = 0$. $\square$

17. Given a differential equation for $y = y(t)$ being:

$$t^3 y'' + t y' - y = 0.$$

- (a) Verify that $y_1(t) = t$ is a solution to the differential equation.
- (b) Find the full set of solutions using reduction of order.
- (c) Show that the set of solutions from part (b) is linearly independent.

Solution:

(a) *Proof.* We note that the left hand side is:

$$t^3 y_1'' + t y_1' - y_1 = t^3 \cdot 0 + t \cdot 1 - t = t - t = 0.$$

Hence $y_1(t) = t$ is a solution to the differential equation. □

(b) By reduction of order, we assume that the second solution is $y_2(t) = tu(t)$, then we plug $y_2(t)$ into the equation to get:

$$2t^3 u'(t) + t^4 u''(t) + tu(t) + t^2 u'(t) = t^4 u''(t) + (2t^3 + t^2)u'(t) = 0.$$

Here, we let $\omega(t) = u'(t)$ to get a first order differential equation:

$$t^2 \omega'(t) = (-2t - 1)\omega(t).$$

Clearly, this is separable, and we get that:

$$\frac{\omega'(t)}{\omega(t)} = -\frac{2t+1}{t^2} = -\frac{2}{t} - \frac{1}{t^2},$$

which by integration, we have obtained that:

$$\log(\omega(t)) = -2\log t + \frac{1}{t} + C.$$

By taking exponentials on both sides, we have:

$$\omega(t) = \exp\left(-2\log t + \frac{1}{t} + C\right) = \tilde{C}e^{1/t} \cdot \frac{1}{t^2}.$$

Recall that we want $u(t)$ instead of $\omega(t)$, so we have:

$$u(t) = \int \omega(t)dt = \tilde{C} \int e^{1/t} \cdot \frac{1}{t^2} dt = -\tilde{C}e^{1/t} + D.$$

By multiplying t , we obtain that:

$$y_2 = -\tilde{C}te^{1/t} + Dt,$$

where Dt is repetitive in y_1 , so we get:

$$y(t) = \boxed{C_1 t + C_2 t e^{1/t}}.$$

(c) *Proof.* We calculate Wronskian as:

$$W[t, te^{1/t}] = \det \begin{pmatrix} t & te^{1/t} \\ 1 & e^{1/t} - \frac{e^{1/t}}{t} \end{pmatrix} = -e^{1/t} \neq 0,$$

hence the set of solutions is linearly independent. □

18. Solve the following second order differential equations for $y = y(x)$:

(a) $y'' + y' - 132y = 0.$

(b) $y'' - 4y' = -4y.$

(c) $y'' - 2y' + 3y = 0.$

Solution:

(a) We find the characteristic polynomial as $r^2 + r - 132 = 0$, which can be trivially factorized into:

$$(r - 11)(r + 12) = 0,$$

so with roots $r_1 = 11$ and $r_2 = -12$, we have the general solution as:

$$y(x) = \boxed{C_1 e^{11x} + C_2 e^{-12x}}.$$

(b) We turn the equation to the standard form:

$$y'' - 4y' + 4 = 0.$$

We find the characteristic polynomial as $r^2 - 4r + 4 = 0$, which can be immediately factorized into:

$$(r - 2)^2 = 0,$$

so with roots $r_1 = r_2 = 2$ (repeated roots), we have the general solution as:

$$y(x) = \boxed{C_1 e^{2x} + C_2 x e^{2x}}.$$

(c) We find the characteristic polynomial as $r^2 - 2r + 3 = 0$, which the quadratic formula gives:

$$r = \frac{2 \pm \sqrt{2^2 - 4 \times 3}}{2} = 1 \pm i\sqrt{2}$$

so with roots $r_1 = 1 + i\sqrt{2}$ and $r_2 = 1 - i\sqrt{2}$, we would have the solution:

$$y(x) = C_1 e^{(1+i\sqrt{2})x} + C_2 e^{(1-i\sqrt{2})x}.$$

To obtain real solution, we apply Euler's identity:

$$y_1(x) = e^x (\cos(\sqrt{2}x) - i \sin(\sqrt{2}x)) \text{ and } y_2(x) = e^x (\cos(-\sqrt{2}x) - i \sin(-\sqrt{2}x)).$$

By the *principle of superposition*, we can linearly combine the solutions to be different solutions, so we have:

$$\tilde{y}_1(x) = \frac{1}{2}(y_1 + y_2) = e^x \cos(\sqrt{2}x),$$

$$\tilde{y}_2(x) = \frac{1}{2}(y_2 - y_1) = e^x \sin(\sqrt{2}x).$$

One can verify that $\tilde{y}_1$ and $\tilde{y}_2$ are linearly independent by taking Wronskian, i.e.:

$$\begin{aligned} W[\tilde{y}_1, \tilde{y}_2] &= \det \begin{pmatrix} e^x \cos(\sqrt{2}x) & e^x \sin(\sqrt{2}x) \\ e^x \cos(\sqrt{2}x) - \sqrt{2}e^x \sin(\sqrt{2}x) & e^x \sin(\sqrt{2}x) + \sqrt{2}e^x \cos(\sqrt{2}x) \end{pmatrix} \\ &= \sqrt{2}e^{2x} \cos^2(\sqrt{2}x) + \sqrt{2}e^{2x} \sin^2(\sqrt{2}x) = \sqrt{2}e^{2x} \neq 0. \end{aligned}$$

Now, they are linearly independent, so we have the general solution as:

$$y(x) = \boxed{C_1 e^x \cos(\sqrt{2}x) + C_2 e^x \sin(\sqrt{2}x)}.$$

19.* Given the following second order initial value problem:

$$\begin{cases} \frac{d^2y}{dx^2} + \cos(1-x)y = x^2 - 2x + 1, \\ y(1) = 1, \\ \frac{dy}{dx}(1) = 0. \end{cases}$$

Prove that the solution $y(x)$ is symmetric about $x = 1$, i.e., satisfying that $y(x) = y(2-x)$.

Hint: Consider the interval in which the solution is unique.

Solution:

Note that I deliberately messed up with all the messy functions. Not only haven't I found a solution to the system, Wolfram cannot have an elementary solution as well. Hence, we need to think, alternatively, on some theorems.

Proof. Here, we suppose that $y(x)$ is a solution, and we want to show that $y(2-x)$ is also a solution.

First we note that we can think of taking the derivatives of $y(2-x)$, by the chain rule:

$$\begin{aligned} \frac{d}{dx}[y(2-x)] &= -y'(2-x), \\ \frac{d^2}{dx^2}[y(2-x)] &= y''(2-x). \end{aligned}$$

Now, if we plug in $y(2-x)$ into the system of equations, we have:

- First, for the differential equation, we have:

$$\begin{aligned} \frac{d^2}{dx^2}[y(2-x)] + \cos(1-x)y(2-x) &= y''(2-x) + \cos(1-x)y(2-x) \\ &= y''(2-x) + \cos(1-(2-x))y(2-x) \\ &= y''(x) + \cos(1-x)y(x) \\ &= x^2 - 2x + 1 = (x-1)^2 = (1-x)^2 \\ &= ((2-x)-1)^2 = (2-x)^2 - 2(2-x) + 1. \end{aligned}$$

- For the initial conditions, we trivially have that:

$$y(1) = y(2-1) \text{ and } y'(1) = y'(2-1).$$

Hence, we have shown that $y(2-x)$ is a solution if $y(x)$ is a solution.

Again, we observe the original initial value problem that:

$$\cos(1-x) \text{ and } x^2 - 2x + 1 \text{ are continuous on } \mathbb{R}.$$

Therefore, by the *existence and uniqueness theorem for second order linear case*, there could be only one solution, which forces that:

$$y(x) = y(2-x),$$

so the solution is symmetric about $x = 1$, as desired. □